PARISHRAM 2.0

CLASS 12TH 2026

Electric Charges and Fields

PHYSICS LECTURE-6

BY-DEV AGNANI SIR



Topics to be covered:

1. Electric Field due to charged Ring on the axis
2. Electric Field due to a Uniformly charged straight rod
3. Electric Lines of Force and its characteristics

imp.

Recap

1. Comparing Electrostatic Force with Gravitational Force
2. Motion of a charged particle in a Uniform Electric Field
3. Electric Field of a Continuous charge distribution
4. Electric Field of a Continuous charge distribution

Continous Charge Distribution :-

① Linear Charge Distribution [1-D]

$$\lambda = \frac{Q}{L} = \frac{dq}{dl}$$

↳ linear charge density

ex: Straight rod, wire.

② Surface charge Distribution [2-D]

$$\sigma = \frac{Q}{A} = \frac{dq}{dA}$$

↳ Surface charge density.

ex: Plane sheet, disk, ring.

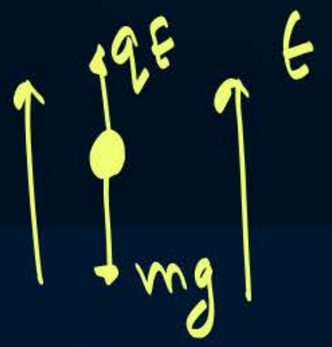
③ Volumetric Charge Distribution [3-D]

$$\rho = \frac{Q}{V} = \frac{dq}{dV}$$

↳ Volumetric charge density.

ex: Sphere, cylinder, cube etc.

QUESTION



The magnitude of electric field intensity E is such that an electron placed in it would experience an electrical force equal to its weight is given by:

A mge

B $\frac{mg}{e}$

C $\frac{e}{mg}$

D $\frac{e^2}{m^2}g$

$mg = qE$

$\Rightarrow E = \frac{mg}{q}$ ← e

$\Rightarrow E = \frac{mg}{e}$



QUESTION

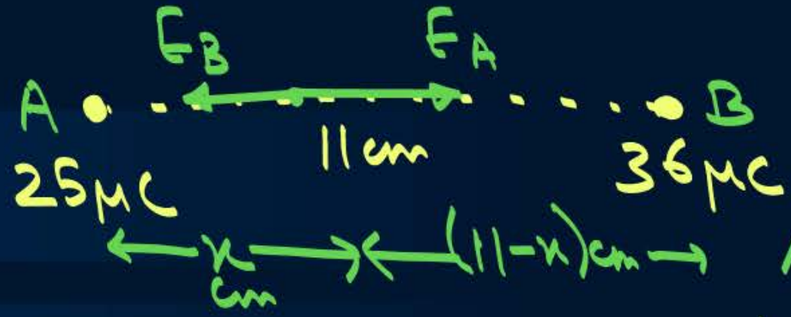
The distance between the two charges 25μC and 36μC is 11 cm . At what point on the line joining the two, the intensity will be zero?

A. At a distance of 5 cm from 25μC

B. At a distance of 5 cm from 36μC

C. At a distance of 10 cm from 25μC

D. At a distance of 11 cm from 36μC



$$E_A = E_B$$

$$\Rightarrow \frac{k(25\mu C)}{x^2} = \frac{k(36\mu C)}{(11-x)^2}$$

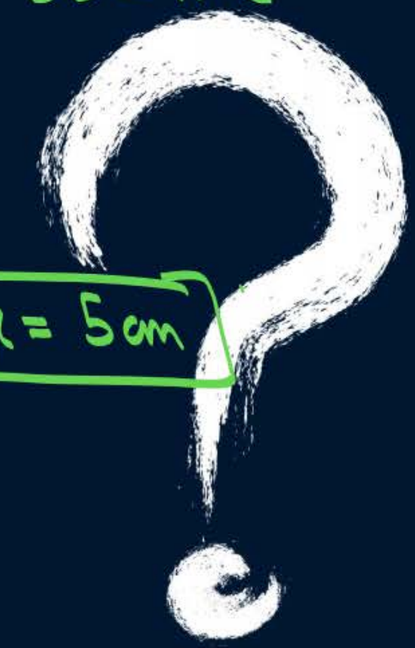
$$\Rightarrow \left(\frac{11-x}{x}\right)^2 = \left(\frac{6}{5}\right)$$

$$\Rightarrow (11-x)5 = 6x$$

$$\Rightarrow 55 - 5x = 6x$$

$$\Rightarrow 55 = 11x$$

$$\Rightarrow x = 5\,cm$$



QUESTION

A charge produces an electric field of 1 N/C at a distance 0.1 m from it. The magnitude of charge is;

$0.1 = 10^{-1}$ m

$E = \frac{kQ}{r^2} \Rightarrow 1 = \frac{9 \times 10^9 \times Q}{10^{-2}\ m^2}$

$\Rightarrow Q = \frac{10^{-2}}{9 \times 10^9}$ C

$\Rightarrow Q = \frac{1}{9} \times 10^{-11}$ C

$\Rightarrow Q = 0.111 \times 10^{-11}$ C

$\Rightarrow \boxed{Q = 1.11 \times 10^{-12}\ C}$

A 1.11×10^{-12} C

B 9.11×10^{-12} C

C 7.11×10^{-6} C

D 6.11×10^{-14} C



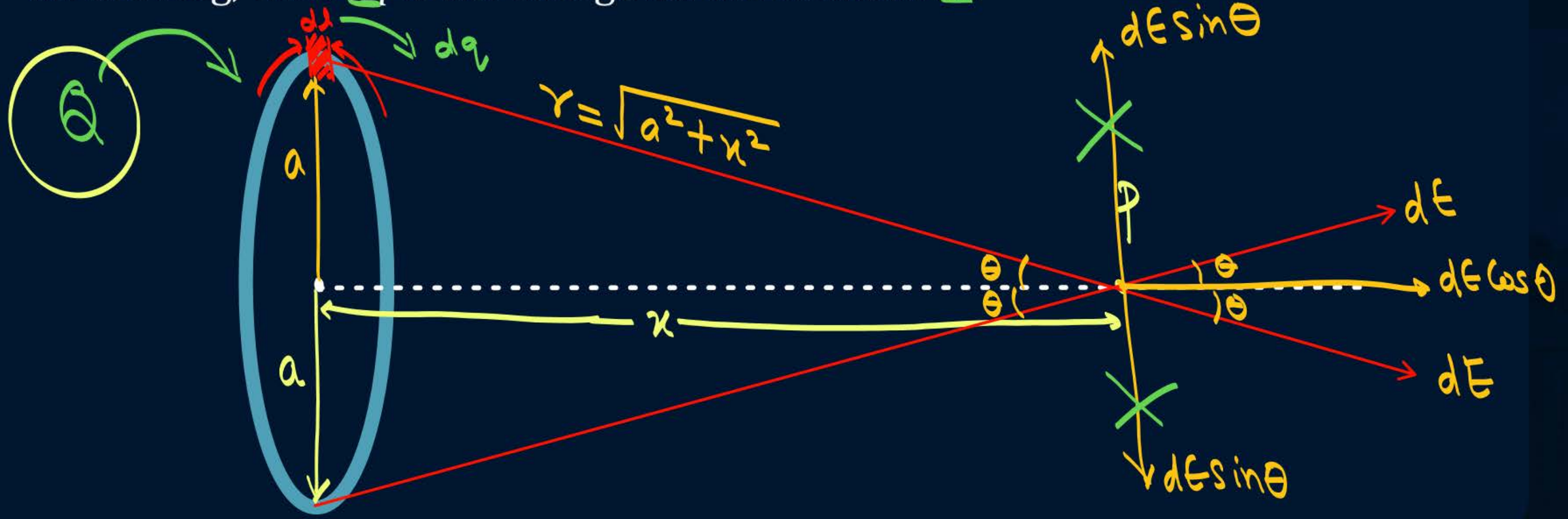
Parishram 2.0 students after Dev sir promised a Friday class, but didn't show up:



Electric Field due to charged Ring on the axis

State Board में नहीं आएगा!

- Suppose that there is a ring lying in the plane perpendicular to the x-axis, and *dl* is the small element on the ring. If the total charge Q is uniformly distributed on the ring, then *dq* is the charge on the element *dl*.



Total electric field due to ring :

$$E = \int \underbrace{dE}_{\text{E.F. due to small charge } dq} \cos\theta$$

$$\Rightarrow E = \int \frac{k\,dq}{r^2}\cos\theta \qquad \left[\cos\theta = \frac{x}{\sqrt{x^2+a^2}}\right] \left[r = \sqrt{x^2+a^2}\right]$$

$$\Rightarrow E = \int \frac{k\,dq}{(x^2+a^2)} \cdot \frac{x}{\sqrt{x^2+a^2}} \Rightarrow E = \frac{kx}{(x^2+a^2)^{1+1/2}} \int dq \to Q$$

$$\Rightarrow \boxed{E = \frac{kQx}{(x^2+a^2)^{3/2}}}$$

; where x = distance of the point on the axis from the centre of the ring where we need to find E.F.

Special Cases :-

(1) Ring acts as a point charge.

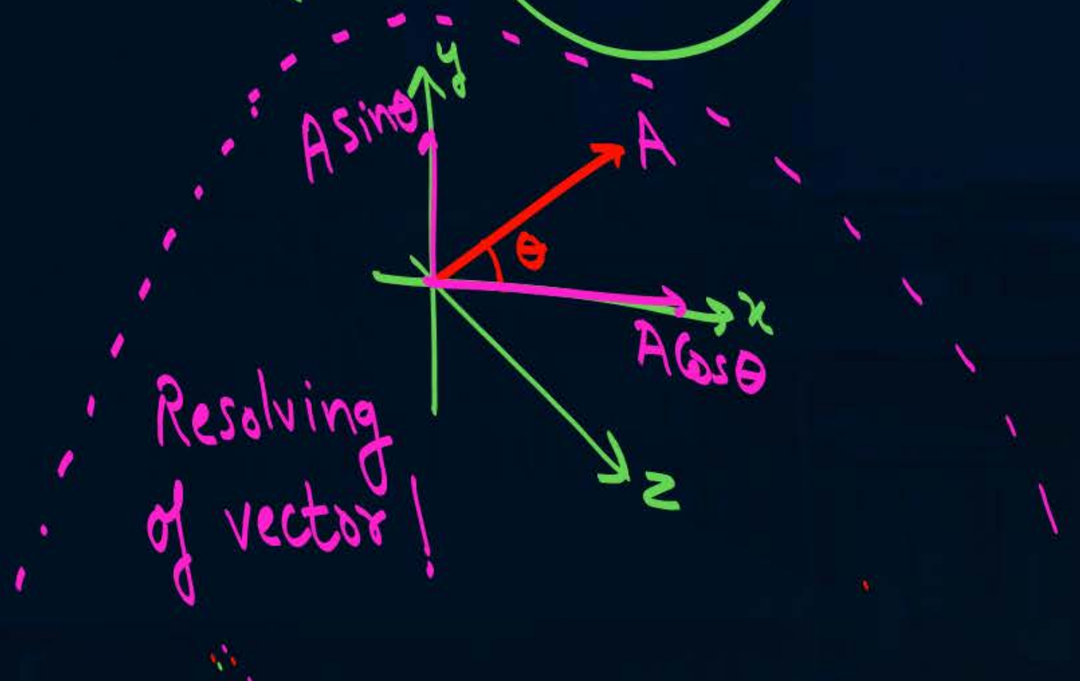
$x >> a$ ⟶ $x^2 >>>>> a^2$

$$E = \frac{kQx}{(x^2+a^2)^{3/2}} \Rightarrow E = \frac{kQx}{(x^2)^{3/2}} = \frac{kQx}{x^3 x^2} \Rightarrow \boxed{E = \frac{kQ}{x^2}}$$

(2) Electric Field at the centre of the ring:

↳ $x = 0$

$$E = \frac{kQx}{(x^2+a^2)^{3/2}} \Rightarrow \boxed{E = 0}$$



QUESTION

Halwaest

A uniformly charged ring of radius R carries a total charge Q. The electric field is measured at a point on the axis of the ring at a distance $x = R$ from its center. Which of the following is the correct value of the electric field?

A $\frac{kQ}{2\sqrt{2R^2}}$

B $\frac{kQ}{4R^2}$

C $\frac{kQ}{3\sqrt{3}R^2}$

D $\frac{kQ}{\sqrt{2}R^2}$

$$E = \frac{KQx}{(x^2+a^2)^{3/2}}$$

$$\Rightarrow E = \frac{KQ(R)}{(2R^2)^{3/2}} = \frac{KQR}{2\sqrt{2}R^3} = \boxed{\frac{KQ}{2\sqrt{2}R^2}}$$



QUESTION

A uniformly charged ring of radius 0.1 m carries a charge of 2×10^{-6} C. The electric field at a point on its axis at a distance 0.1 m from the center will be:
($k = 9 \times 10^9\ Nm^2/C^2$)

A 3.18×10^5 N/C

B 2.25×10^5 N/C

C 4.50×10^5 N/C

D 6.36×10^5 N/C

$$E = \frac{kQx}{(x^2+a^2)^{3/2}} = \frac{(9\times10^9)(2\times10^{-6})(10^{-1})}{(10^{-2}+10^{-2})^{3/2}}$$

$$E = \frac{18\times10^2}{2\sqrt{2}\times10^{-3}} = \frac{9}{\sqrt{2}}\times10^5$$

$$= \frac{90}{1.4}\times10^5$$



INTEGRATION :-

$$\int dy = \int (x^5)\, dx$$

$$\Rightarrow y = \left(\frac{x^{5+1}}{5+1}\right) = \frac{x^6}{6}$$

$$\therefore y = \int \left(\frac{1}{x^2}\right) dx$$

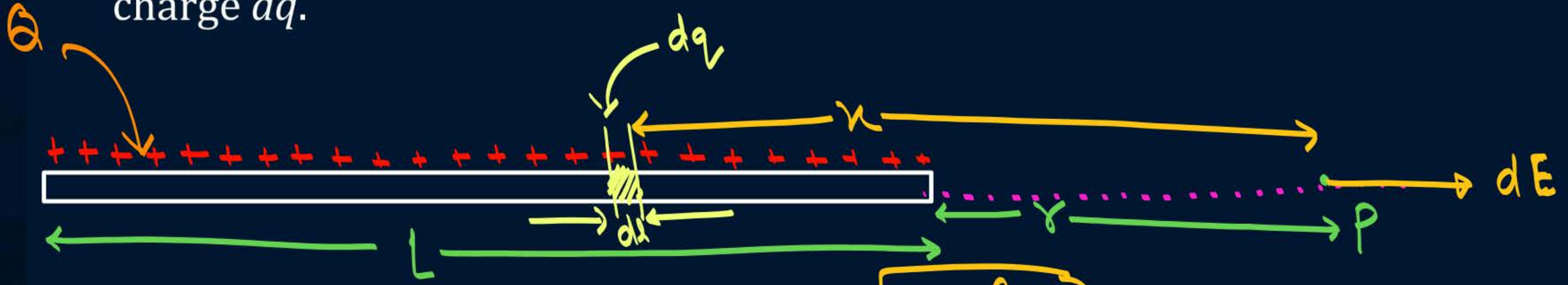
$$\Rightarrow y = \left(\frac{x^{-2+1}}{-2+1}\right)$$

$$\Rightarrow y = \left(\frac{x^{-1}}{(-1)}\right)$$

$$\Rightarrow y = \left(\frac{-1}{x}\right)$$

Electric Field due to a Uniformly charged straight rod

- Suppose we have a Uniformly charged straight rod of length L, with total charge Q. *dE* will be the electric field at point P lying on the axis of the rod at distance r from the rod, due to the small element *dl* on the rod having the charge *dq*.



Total electric field at point P due to charged straight rod. $= E = \int dE$

$$E = \int dE = \int \frac{k\,dq}{x^2}$$

$$\lambda = \frac{Q}{L} = \frac{dq}{dx} \Rightarrow \boxed{dq = \frac{Q}{L}\,dx}$$

$$\Rightarrow E = \int_{x=r}^{x=r+L} \frac{k\,Q}{L}\,\frac{dx}{x^2}$$

$$\Rightarrow E = \frac{kQ}{L}\int_{x=r}^{x=r+L} \frac{dx}{x^2} \Rightarrow E = \frac{kQ}{L}\left[\frac{x^{-2+1}}{-2+1}\right]_{x=r}^{x=r+L}$$

$$\Rightarrow E = \frac{kQ}{L}\left[\frac{-1}{x}\right]_{x=r}^{x=r+L} \Rightarrow E = \frac{kQ}{L}\left[\frac{-1}{(r+L)} - \left(\frac{-1}{r}\right)\right]$$

$$E = \frac{KQ}{L}\left[\frac{1}{r} - \frac{1}{r+L}\right]$$

$$\Rightarrow E = \frac{KQ}{L}\left[\frac{r+L-r}{(r)(r+L)}\right]$$

$$\Rightarrow E = \frac{KQ}{L}\left[\frac{L}{(r)(r+L)}\right]$$

$$\Rightarrow \boxed{E = \frac{KQ}{(r)(r+L)}}$$

Here, r is the distance of the rod from point P.

QUESTION

A uniformly charged rod of length 0.4 m has a total charge 6×10^{-6} C. The electric field at a point along its axis 0.2 m from one end of the rod:
($k = 9 \times 10^9$ Nm^2/C^2)

$E = \frac{kQ}{(r)(r+L)}$

$\Rightarrow E = \frac{(9 \times 10^9)(6 \times 10^{-6})}{(0.2)(0.4)}$

$\Rightarrow E = \frac{9 \times 10^9 \times 10^5 \times 10^{-5}}{2 \times 10^{-1}}$

$\Rightarrow E = 9 \times 5 \times 10^4$

$\Rightarrow E = 4.5 \times 10^5$ N/C

A 4.50×10^5 N/C

B 3.38×10^5 N/C

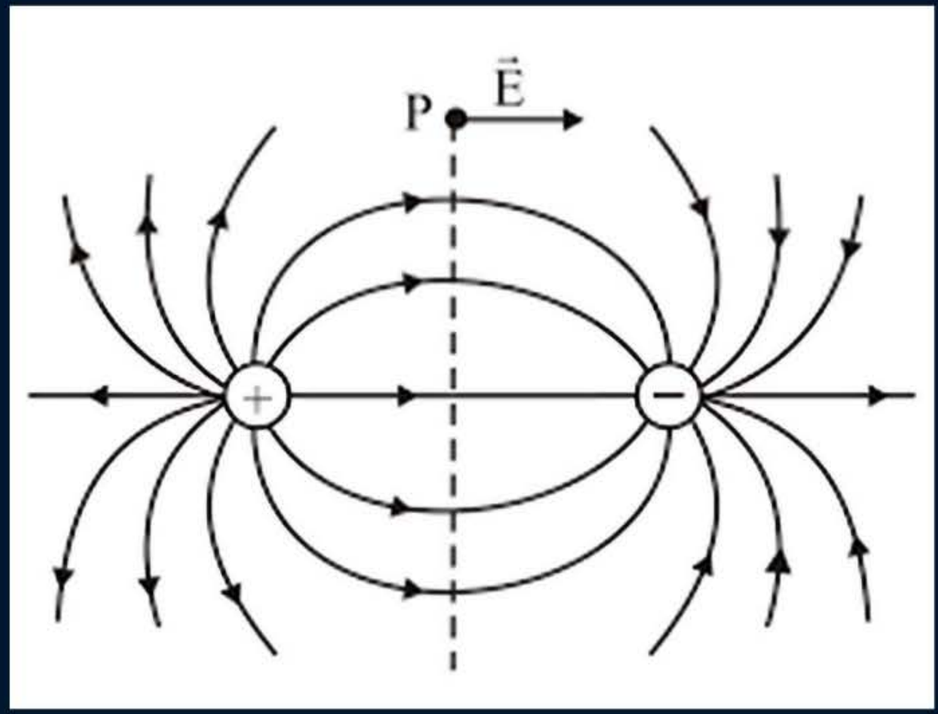
C 4.50×10^5 N/C

D 5.06×10^5 N/C



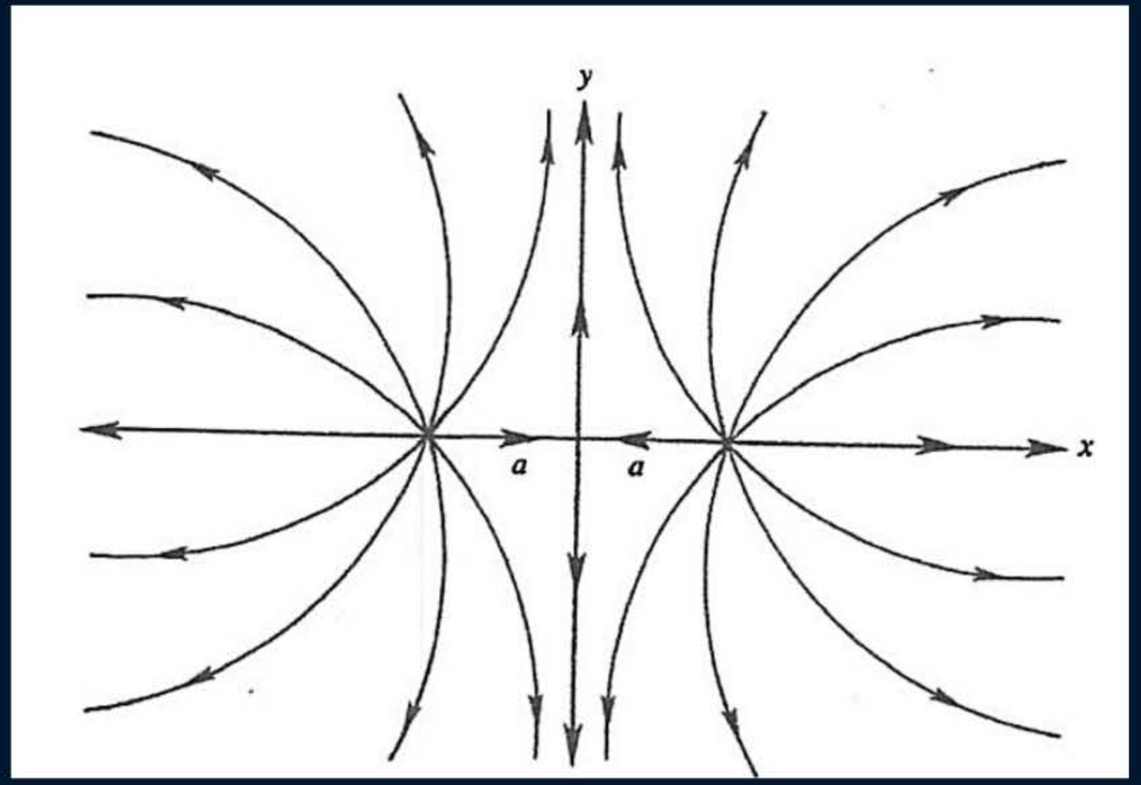
Electric Lines of Force and its characteristics

- Electric line of force is an imaginary path along which a unit +ve test charge would tend to move in an electric field. Electric lines of force start from +ve charge and end at –ve charge.



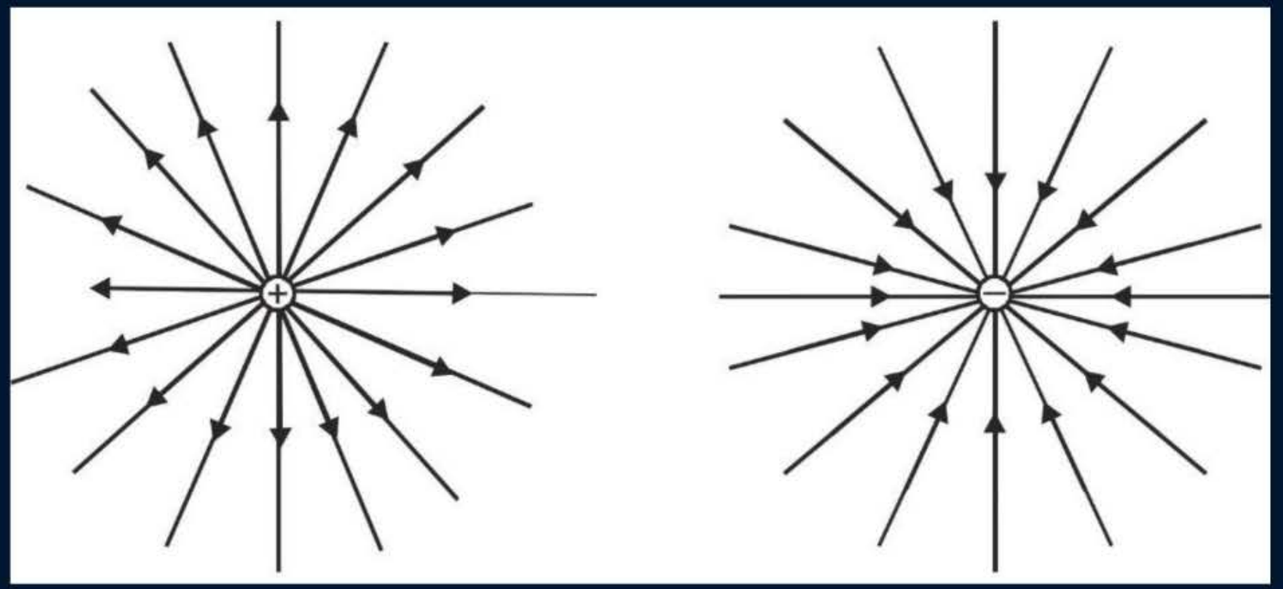
Electric Lines of Force and its characteristics

- Number of electric field lines originated from positive charge or terminated to negative charge is directly proportional to strength of charge.



Electric Lines of Force and its characteristics

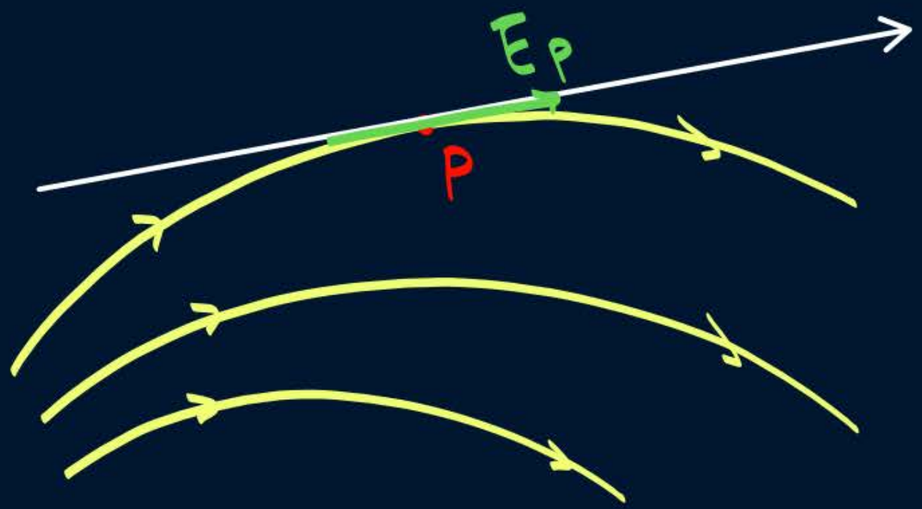
- Electric lines of force in the case of isolated +ve charge are radially outwards and in the case of isolated-ve charge are radially inwards.



- $|E_A| > |E_B|$ as density of field lines is greater than at B.

Electric Lines of Force and its characteristics

- The tangent at any point to the curve gives the direction of electric field at that point.



- Electric lines of force do not intersect.
- Electric lines of force never exist inside a conductor.
- The force experienced by a unit positive test charge placed at a point in the electric field gives the intensity of electric field at that point both in magnitude and direction.

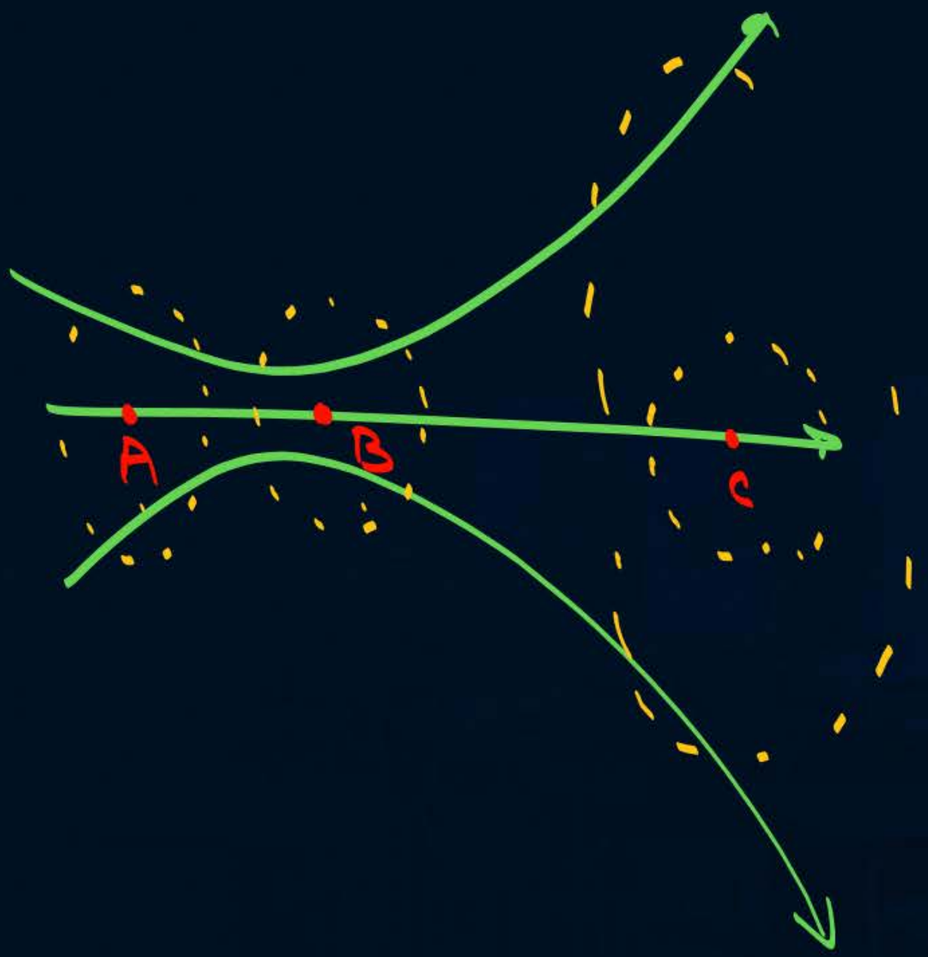
Que: Tick the correct option on the basis of density of E.F. ?

A) $E_A > E_B > E_C$

B) $E_B > E_A > E_C$

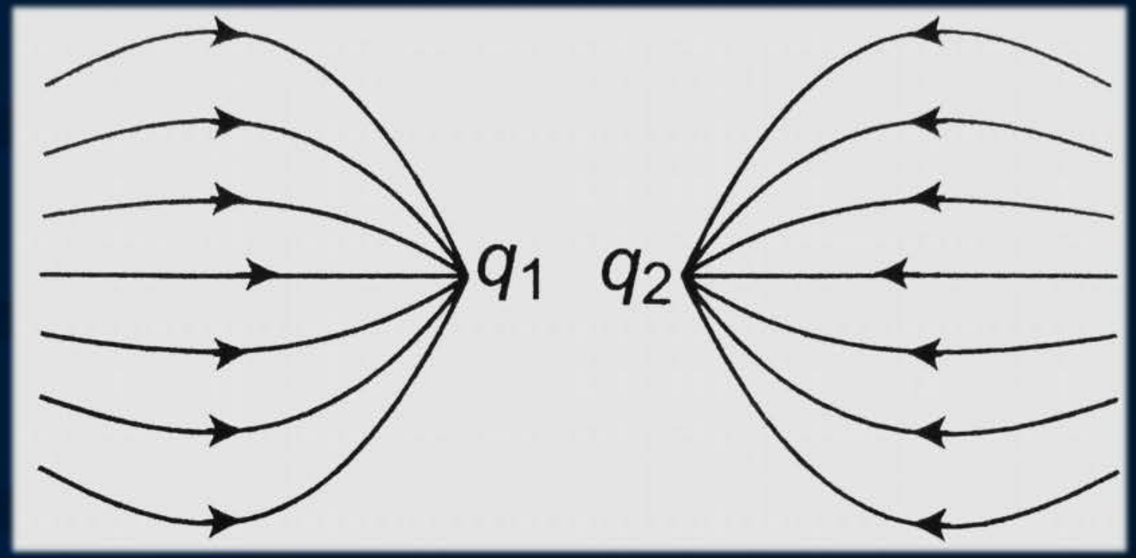
C) $E_B > E_C > E_A$

D) $E_C > E_A > E_B$



QUESTION

The given figure gives electric lines of force due to two charges q_1 and q_2. What are the signs of the two charges?



A q_1 is positive but q_2 is negative

B q_1 is negative but q_2 is positive

C Both are negative

D Both are positive

QUESTION

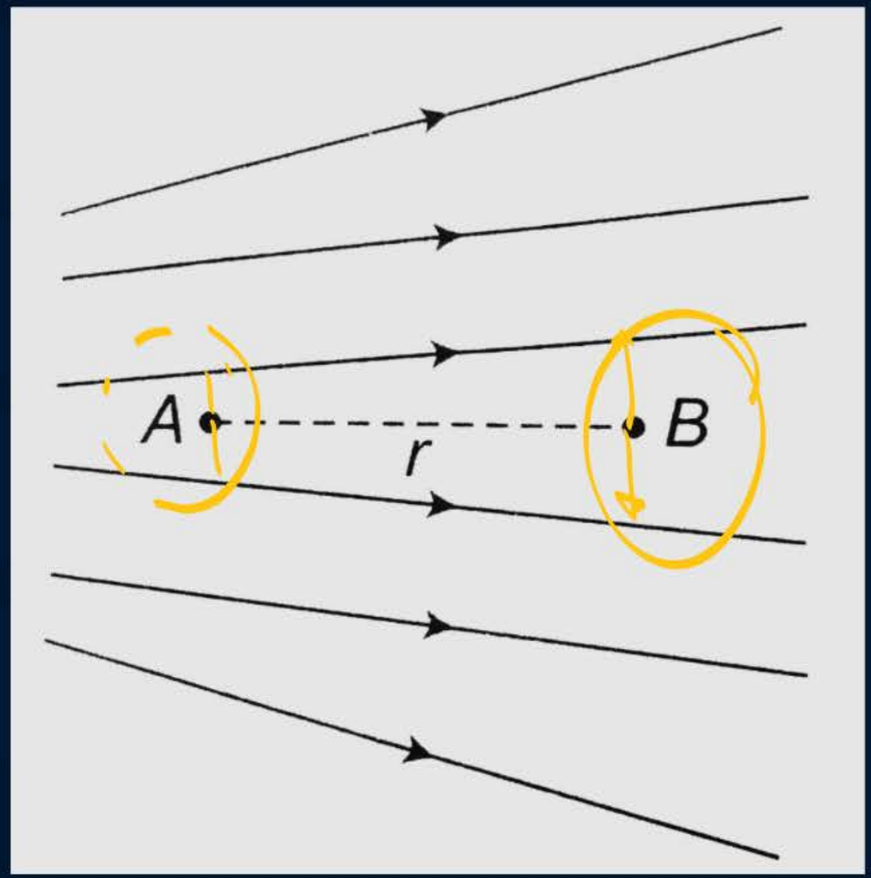
Figure shows the electric lines of force emerging from a charged body. If electric field at A and B are E_A and E_B respectively and distance between A and B is r then

A $E_A > E_B$

B $E_A = E_B$

C $E_A < E_B$

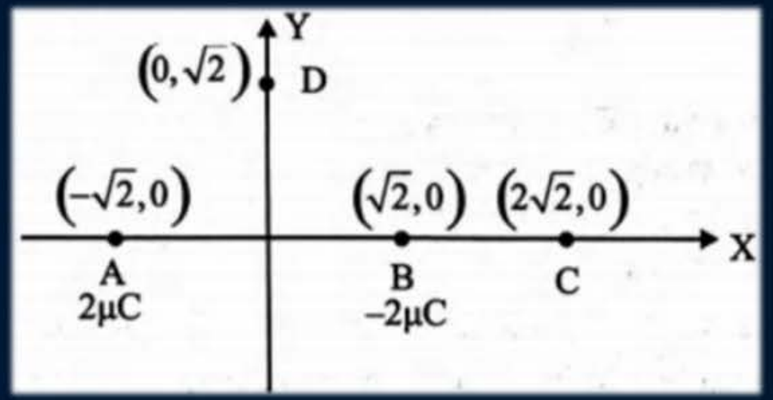
D $E_A > \frac{E_B}{r^2}$



QUESTION

HW

Two point charges 2 μC and -2 μC are placed at point A and B as shown in figure. Find out electric field intensity at point C (All the distances are measured in meter.)



A $-8000\hat{i}N/C$

B $-800\hat{i}N/C$

C $800\hat{i}N/C$

D $+8000\hat{i}N/C$

QUESTION

$F = qE$

$ma = qE \rightarrow \left(a = \frac{qE}{m}\right)$

$u = 0$

An electron of mass m and charge q is accelerated from rest in a uniform electric field of strength E. The velocity acquired by it as it travels a distance l is

$v = ?$

A $\sqrt{\frac{2Eql}{m}}$

B $\sqrt{\frac{2Eq}{m}}$

C $\sqrt{\frac{2Em}{ql}}$

D $\sqrt{\frac{Eq}{ml}}$

$2as = v^2 - u^2$

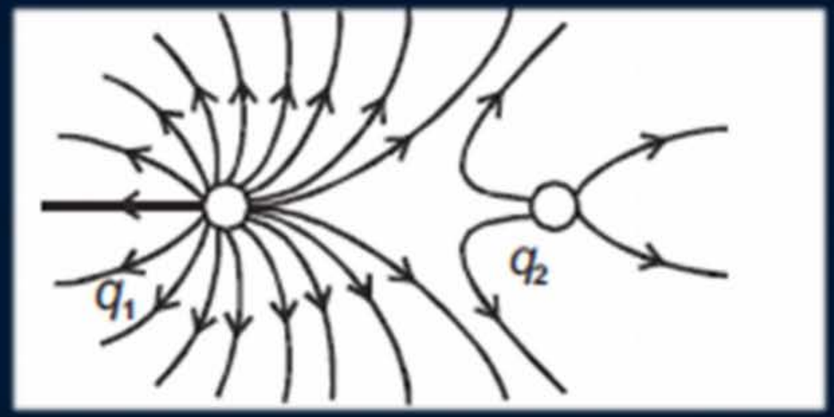
$\Rightarrow 2\left(\frac{qE}{m}\right)l = v^2 - 0$

$\Rightarrow \boxed{v = \sqrt{\frac{2qEl}{m}}}$



QUESTION

Figure shows electric field lines due to a charge configuration from this we conclude that



A q_1 and q_2 are positive and $q_2 > q_1$

B q_1 and q_2 are positive and $q_1 > q_2$

C q_1 and q_2 are negative and $|q_1| > |q_2|$

D q_1 and q_2 are negative and $|q_2| > |q_1|$

मज़ा ∫ (L)
× (D)

THANK YOU

